

v.	$x_{it} = r_{it} - r_{it+1}$	Trade Schedule
vi.	$\alpha_{i,\min}^* \leq \frac{x_{it}}{v_{it}} \leq \alpha_{i,\min}^*$	Trade Rate Bounds
vii.	$x_{i,\min}^* \leq x_{it} \leq x_{i,\max}^*$	Trade Size Bounds
viii.	$r_{i,\min}^* \leq r_{it} \leq r_{i,\max}^*$	Residual Size Bounds
ix.	$LB \leq \sum_{i=1}^m \sum_{j=1}^k (side(i) \cdot x_{ij} \cdot p_{ik}) \leq UB$	Self-Financing
x.	$LB \leq (side(i) \cdot r_{ij} p_{ij}) \leq UB$	Risk-Management

Constraint Description

Investors may include all or some of the constraints above. These constraints are fund specific and can be omitted if deemed unnecessary by the trader. These constraints are described as follows:

- i. **Completion:** ensures that the optimization solution will executes all shares in all orders within the defined trading horizon.
- ii. **No Short Sales:** ensures that the side of the order will not change. For example, the optimization solution will only buy shares for a buy order and sell shares for a sell order. Without this constraint, it is possible that the optimization may overbuy or oversell during the day and then have to offset the newly acquired position.
- iii. **Shrinking Portfolio:** ensures that the size of the order keeps decreasing. For example, if the order is to buy 100,000 shares the positions will always be decreasing towards zero and will never increase. Without this constraint the optimization solution may determine it would be best to first sell 25,000 shares so that the order increases to 125,000 shares. While this type of strategy may be the best way to manage overall portfolio risk it may not be an acceptable solution for the investor. For example, it exposes the investor to short-term risk if the stock is halted after the position size increases to 125,000 shares.
- iv. **Residual Schedule:** defines the residual share quantity in each period in terms of unexecuted shares at that point in time. Used if the decision variable is the trade share amount.
- v. **Trade Schedule:** defines the shares to trade in terms of the residual shares. Used if the decision variable is the residual trade vector.
- vi. **Trade Rate Bounds:** the defined maximum and minimum trading rates. For example, investors may wish to trade at least 1% of the total market volume in each period but no more than say 25% of total market volume in each period. These constraints are most often defined in terms of percentage of volume rate so may need to be converted to the trade rate definition.